INCH-POUND

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SUPERSEDING
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MILITARY SPECIFICATION

FOOTWEAR COVERS, TOXICOLOGICAL AGENTS PROTECTIVE, M-1

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of M-1 toxicological agents protective footwear covers.

1.2 Classification. The footwear cover shall be in the following sizes as specified (see 6.2).

Schedule of Sizes

Small
Medium
Large

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8430

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

A-A-50083	- Bag, Plastic, Folded Garment
V-T-285	- Thread, Polyester
CCC-C-419	- Cloth, Duck, Cotton, Unbleached, Plied-Yarns Army and Numbered
CCC-D-950	- Dyeing and Aftertreating Processes for Cotton Cloths
DDD-L-20	- Label: For Clothing, Equipage, and Tentage, (General Use)
PPP-B-636	- Boxes, Shipping, Fiberboard
PPP-T-45	- Tape, Gummed, Paper, Reinforced and Plain, for Sealing and Securing

MILITARY

MIL-W-530	- Webbing, Textile, Cotton, General Purpose, Natural or in Colors
MIL-C-12189	- Cloth, Coated: Butyl Coated, Toxicological Agents Protective
MIL-L-35078	- Loads, Unit: Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment and Equipage; General Specification For
MIL-C-43734	- Cloth, Duck, Textured Nylon

STANDARDS

FEDERAL

FED-STD-191	- Textile Test Methods
FED-STD-751	- Stitches, Seams, and Stitchings

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	- Marking for Shipment and Storage

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Stitching X

MIL-STD-147 - Palletized Unit Loads
MIL-STD-731 - Quality of Wood Members for Containers and Pallets

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

THE COLOR ASSOCIATION OF THE UNITED STATES, INC.

Standard Color Card of America

Department of Defense Standard Shades for Sewing Threads

(Color cards may be available from the Color Association of the United States, Inc., 343 Lexington Avenue, New York, NY 10016-0927. If color cards are not available from the Color Association, individual color samples may be obtained from the contracting activity or as directed by the contracting activity.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 Samples. Standard samples, when furnished, are solely for guidance and information to the contractor (see 6.4). Variation from this specification may appear in the sample, in which case this specification shall govern.

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3.3 Material. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification.

3.3.1 Coated cloth. The coated cloth for the panel and inner sole shall be butyl coated cloth conforming to MIL-C-12189 except that the resistance to mustard (H), initial, after weatherometer, and after decontamination shall be 50 minutes (minimum).

3.3.2 Duck cloth. The duck cloth used in the outer sole shall be cotton conforming to type I, No. 10, hard texture of CCC-C-419 or nylon conforming to class 1 of MIL-C-43734.

3.3.2.1 Dyeing and aftertreatment. The duck shall be dyed Olive Drab 7 and aftertreated (cotton duck only) as specified for type I, class B, inhibitor (d) of CCC-D-950.

3.3.3 Tie tapes. The tie tapes shall be type IIa, class 7 webbing, 3/4 inch wide, dyed Olive Drab 7 conforming to MIL-W-530.

3.3.4 Thread. The thread shall conform to type I, class 1, sub-class A, sizes B and E of V-T-285 except that colorfastness requirements shall not apply. The thread shall be dyed Olive Drab S-1 (C.A. 66022).

3.3.5 Strapping. The strapping used to seal the seams shall be made of the coated cloth specified in 3.3.1, straight or bias cut, except that a cotton-base cloth may be used in place of nylon provided the finished coated cloth meets the requirements of MIL-C-12189. The strapping used to seal the peripheral seam on the outside of the sole shall be 1-1/2 $\pm$ 1/16 inches wide. The strapping for the back seam and the inside of the sole seam shall be 1 $\pm$ 1/16 inch wide.

3.3.6 Cement. The cement used to adhere the strapping to the seams shall be of the self-curing neoprene or butyl base type. The cement, when applied and tested as specified in 4.4.1, shall not crack or flake when subjected to the cold crack test. The hydrostatic resistance of the cement coated cloth shall be not less than 90 percent of the actual hydrostatic resistance of the butyl coated cloth prior to cementing. Sample strapped seams, prepared with the cement as specified in 4.4.1, shall have a minimum adhesion of 2.0 pounds per inch of width after water immersion and after heat aging.

3.3.7 Labels. Each footwear cover shall have a combination size and identification label that includes the date of manufacture and conforms to type IV or VI, class 1 and class 2 of DDD-L-20, with the following exceptions:

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Type VI label (cloth)

a. The label shall be printed prior to impregnation or coating.
b. Only chlorine bearing vinyl resin composition shall be used for impregnation or coating.
c. The label shall measure 2-3/4 by 1-1/2 inches, and the legend shall be printed parallel to the small dimension.
d. All colorfastness and writing tests are not applicable.

Type IV label (stamped)

a. A vinyl base black indelible ink shall be used.
b. Letters shall be sans-serif, extra bold, monotone style, 14 point (caps only).

3.3.7.1 Colorfastness of labels. Both the cloth label and the stamped label shall be legible and shall not become tacky after testing as specified in 4.4.1.

3.4 Design. The footwear cover shall be a loose fitting cover with two sets of tie tapes as illustrated in figure 1. The tie tapes hold the footwear cover securely over the wearer's footwear.

3.5 Construction.

3.5.1 Patterns. Standard patterns will be furnished by the Government and provide the proper allowance for all seams. Standard patterns shall not be altered in any manner and are to be used only as a guide for cutting the contractor's working patterns.

3.5.2 Manufacturing operations. The footwear cover shall be manufactured by and with the use of all operations specified in 3.5.6. The contractor is not required to follow the exact sequence of operations.

3.5.3 List of pattern parts. The component parts for a pair of footwear covers shall be cut from the material and in the number of parts as specified in table I.

TABLE I. Pattern parts

Material	Item	Nomenclature	Cut parts
Butyl coated cloth	1	Panel cover	2
	2	Sole, inner	4
Cotton duck	1	Sole, outer	2

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3.5.4 Stitches, seams, and stitching. Stitch, seam, and stitching types, as specified in 3.5.6, shall conform to FED-STD-751. Unless otherwise specified, all single row stitching shall be placed 1/16 to 1/8 inch from the outer edge, and all double needle stitching shall be 3/16 to 5/16 inch gage with the outer row of stitching 1/16 to 1/8 inch from the outer edge. All stitching shall have 8 to 10 stitches per inch. Seam allowance shall be maintained and seams properly sewn so that no run-offs, twists, pleats, puckers or open seams will result.

3.5.4.1 Backstitching. Thread breaks in stitching shall be backstitched not less than 1 inch. Ends of seams and stitching shall be backstitched not less than 1 inch except where the ends are caught in other seams or stitching.

3.5.4.2 Precementing. All joining seams except seam for joining sole section to panel shall be overlapped 3/8 $\pm$ 1/16 inch and cemented before stitching.

3.5.5 Application of seam strapping. Strapping shall be applied to both the inside and outside of all seams, using the cement specified in 3.3.6. The strapping shall be applied after all stitching operations are completed, including tacking of the tapes, except that the upper tie tapes shall be bartacked as specified in 3.5.6.5 after application of the strapping to the outside of the back seam. Each individual seam shall be strapped with a continuous length of strapping. Where ends of strapping meet, a 1 inch (+1/2, -1/4 inch) overlap shall be made. Wherever strappings intersect, they shall be overlapped. When the strapping is centered over the seam and pressed into position, the cement line shall not extend more than 1/2 inch on either side of the strapping.

3.5.5.1 Preparation of seam area for strapping. Prior to application of the cement, the seam and area along the seam where strapping is to make contact and the surface of the strapping to be adhered shall be thoroughly cleaned with a suitable cleaning agent that is not a solvent for the butyl coating. In addition, the preceding operation shall be employed for all cementing and precementing operations.

3.5.6 Stitching operations.

3.5.6.1 Assembly of the footwear cover. The footwear cover shall be constructed and assembled with the heavily coated side of the coated cloth, including strapping on the outside. Size E thread shall be used for all operations except for stitching the turned ends of the tie tapes and stitching of label, where size B thread may be used.

3.5.6.2 Panel section. The panel section shall be made of a double thickness of coated cloth folded from a single piece with a single seam along the back. A 3/8 $\pm$ 1/16 inch cut shall be made at the fold on one side of the panel section such that the edges can be alternately lapped and cemented together. The back seam shall be precemented and sewn using seam type LSa-1 and stitch type 301, 3/16 to 5/16 inch from the edge.

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3.5.6.3 Sole section. The sole section shall be made of two coated cloth innersoles and a single cotton duck outersole stitched together using seam type SSa-1 and stitch type 301.

3.5.6.4 Cover assembly. The sole section shall be joined to the panel section using a SSa-2 seam type and stitch type 301. The stitched seam shall be cemented prior to strapping.

3.5.6.5 Upper tie tape. The upper tie tape shall be made in one piece. Each end shall be turned under and, using seam type EFb-1, shall be stitched across with stitch type 301 to securely tack the ends. The tie tape shall be stitched or bartacked at its center point to the back seam of the cover where indicated on the patterns. Seam type SSa-2 and stitch type 301 shall be used for stitching. Bartacking shall be 1/2 $\pm$ 1/16 inch long with a 1/16 inch bight and shall contain a minimum of 28 stitches. The two ends of the tie tapes shall each extend 25 $\pm$ 1/2 inches from the back seam of the cover. The tie tapes shall be stitched or bartacked to the back seam over the strapping in such a manner that the bartacking or stitching is completely covered on the inside of the cover by the strapping, or by a round patch as specified in 3.8.

3.5.6.6 Lower tie tapes. The two lower tie tapes shall be stitched to the outer layer of cover material where indicated on the patterns. A piece of the cotton duck specified in 3.3.2, approximately 2 inches by 2 inches, shall be used to reinforce the stitching. The reinforcement shall be sandwiched between the two layers of cover material. A piece of the coated cloth specified in 3.3.1, approximately 3 inches by 3 inches, shall have one edge caught in the bottom row of stitching and shall be cemented to the inside of the cover to completely cover the tie tape stitching. The stitching used to attach the tapes to the cover shall be a rectangular box of stitching, 1-3/8 inches long by 9/16 inch wide, reinforced top and bottom with an extra row of stitching. The outer ends shall be turned under and stitched using seam type EFb-1 and stitch type 301 to securely tack the ends. Each tie tape shall extend 36 $\pm$ 1/2 inches from the sole seam.

3.5.7 Labeling. The cloth label (when used) shall be stitched on four sides to a piece 3-1/4 by 2 inches of the coated cloth specified in 3.3.1, using stitch type 301. The assembly shall be cemented to the inside of the cover with the cement specified in 3.3.6, so that the upper left hand corner of the label will be 2 $\pm$ 1 inches from the top of the cover and 2 $\pm$ 1 inches to the right of the nearest edge of the back seam strapping. Alternately, the label shall be stamped directly on the inside of the cover and shall be positioned as specified above.

3.6 Measurements. The measurements of the finished footwear cover shall conform to those specified in figure 1.

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3.7 Performance.

3.7.1 Adhesion of strapping. After application, as specified in 3.5.5, the strapping shall have a minimum adhesion of 3.0 pounds per inch of width when tested as specified in 4.4.5.

3.7.2 Leakage. The finished footwear cover shall not leak when tested as specified in 4.4.5.

3.8 Repairs in coating of finished footwear cover. Areas containing coating defects may be repaired by application of a round patch, 1-1/2 ± 1/4 inches in diameter. The patches shall be made of the coated cloth specified in 3.3.1 and shall be adhered with the cement specified in 3.3.6. The lightly coated side of the patch shall be cemented to the footwear cover, and the patch shall extend a minimum of 1/2 inch beyond all edges of the damaged area. Maximum length of defective area shall be 3/4 inch. The patch shall be smoothly and evenly adhered over its entire area and shall be well adhered to the coated cloth so that, when subjected to flexing action between the hands, it will show no lifting of the edge, and no cracking, flaking, or removal of the cement or patch. Patches shall not overlap seams or other patches. The repaired area shall show no stiffening or tackiness that would affect serviceability. Repairing of fabric coating defects by methods other than patching are not authorized.

3.8.1 Location and number of repairs. Repairs not exceeding four in number may be made on any one footwear cover. Repairs may be located on either the outside or the inside coated cloth layer of the footwear cover.

3.8.2 Replacement of defective components. During the cutting and manufacturing process, components having material defects or damages that are classified as defects in 4.4.2.2 and 4.4.3 shall be removed from production and replaced with nondefective and properly matched components.

3.9 Workmanship. The end item shall conform to the quality of product established by this specification. All cemented areas shall be fully cured prior to packaging. The occurance of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for ensuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point, or at all points in the manufacturing process necessary to assure compliance with all dimensional requirements.

4.1.3 Certificates of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 3.1 and 6.2), it shall be examined for the defects specified in 4.4.3 and 4.4.4 and tested as specified in 4.4.5.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document. In addition to testing provisions contained in subsidiary documents, testing shall be performed on components and materials listed in table II. When the data in the "Number of determinations per sample unit" and "Results reported as" columns are not specified in table II, they shall be as required by the referenced test methods. All test reports shall contain the individual values used in expressing the final result. The sample size shall be as follows:

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Lot size	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

The lot shall be unacceptable if one or more sample units, the lot average, or the composite sample fails to meet any of the specified test requirements.

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TABLE II. Component tests

Component and lot expressed in terms of	Characteristic	Requirement paragraph	Test method	Requirements applicable to: Sample unit	Requirements applicable to: Lot avg	No. of determinations per sample unit	Results reported as: Pass or fail	Results reported as: Numerically to nearest	Sample unit
Cement, 5 gallons	Material identification	3.3.6	1/	-	-	-	-	-	-
	Cold crack	3.3.6	4.5.2	X	-	1	X	-	-
	Hydrostatic resistance	3.3.6	4.5.3	-	X	5	-	1 percent	1 pint composite and 1 yard of coated cloth and 2-yard lengths of cemented strapped seam
	Adhesion after water immersion	3.3.6	4.5.1.2 2/	-	X	5	-	0.1 pound per inch of width	
	Adhesion after heat aging	3.3.6	4.5.1.3 2/	-	X	5	-	0.1 pound per inch width	
Label, cloth (when applicable) (1 gross)	Identification of impregnating material	3.3.7	1/	-	-	-	-	-	-

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TABLE II. Component tests - Continued

Component and lot expressed in terms of	Characteristic	Requirement paragraph	Test method	Requirements applicable to: Sample unit	Requirements applicable to: Lot avg	No. of determinations per sample unit	Results reported as: Pass or fail	Results reported as: Numerically to nearest	Sample unit
Label, cloth (continued)	Resistance to fading and tackiness	3.3.7.1	4.5.5	X	-	2	X	-	2 labels
Ink (applicable when marking is stamped directly to end item) 1 gallon	Resistance to fading and tackiness	3.3.7.1	4.5.5	X	-	2	X	-	2 labels 3/

1/ A certificate of compliance shall be submitted and will be acceptable for the stated requirement.

2/ Two 4-inch wide by 1-yard long samples of coated cloth specified in 3.3.1 shall be joined using seam and stitch type specified in 3.5.6.2 for the back seam. The seamed samples shall then be strapped on one side only with a 1-inch strip of strapping material specified in 3.3.5 using the cement specified in 3.3.6.

3/ A piece of the cloth with the legend stamped thereon shall be considered as one label. The coated cloth need only be large enough to include the legend.

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4.4.2 In-process inspection. Inspection of subassemblies shall be made to ascertain that construction details which cannot be examined in the finished product are in accordance with specified requirements. The Government reserves the right to exclude from consideration for acceptance, any material or service for which in-process inspection has indicated nonconformance.

4.4.2.1 Cut part critical defect examination. Every cut part, prior to the in-process examination in 4.4.2.2, shall be examined for the critical defects listed in table III. Any cut part found to contain one or more critical defects shall be rejected.

4.4.2.2 In-process examination of cut parts. The components of the footwear cover shall be examined after cutting and prior to assembly to determine that the contractor is cutting in accordance with patterns and is removing parts containing material defects and damages. Defects found shall be classified as listed in table III. The lot size shall be expressed in terms of panels, innersoles, or outersoles, as applicable. The sample unit shall be one panel, one innersole, or one outersole, as applicable. The inspection level shall be I, and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 1.0 for major defects and 4.0 for total (major and minor combined) defects. Any critical defect found during this examination shall be cause for rejection of the lot.

TABLE III. Cut part defects

Examine	Defect	Classification		
		Critical	Major	Minor
Panel and innersole	Abrasion mark or scratch which does not expose the base fabric		101	
	Cut, hole or tear which exposes or extends through the base fabric	1		
	Uncoated area	2		
	Pit, blister, tunnel, or delamination of coating	3		
	Lump or imbedded foreign matter which protrudes so as to be readily abraded from material or easily removed	4		

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TAbLE III. Cut part defects - Continued

Examine	Defect	Classification		
		Critical	Major	Minor
Panel and innersole (continued)	Crease or wrinkle resulting in adhesion of surface or delamination of coating when corrected by manual pressure	5		
	NOTE: A pick glass may be used to determine degree of deficiency under coated fabric classifications.			
Outersole	Cut, hole, tear, smash, or mend (not extended through to innersole)		102	
	Hard crease or wrinkle			201
	Weakening defects, area with visibly thinner or fewer yarns, float, or loose slubs		103	
	Two or more yarns broken, missing, or not interlaced		104	
	Miscut	6		

4.4.2.3 End item critical defect examination. Every end item, prior to the visual and dimensional examinations in 4.4.3 and 4.4.4, shall be examined for the critical defects listed in table IV. Any end item found to contain one or more critical defects shall be rejected.

4.4.3 End item visual examination. The end items shall be examined for the defects listed in table IV. The lot size shall be expressed in units of footwear covers. The sample unit shall be one footwear cover. The inspection level shall be II and the AQL, expressed in terms of defects per hundred units, shall be 2.5 for major defects and 15.0 for total (major and minor combined) defects. The finding of any critical defect shall be cause for rejection of the lot.

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TABLE IV. End item visual defects

Examine	Defect	Classification		
		Critical	Major	Minor
Cleanness	Spot, stain, or streak more than 1 inch in combined directions that cannot be readily removed with naptha or water			201
	Cement line extends more than 1/2 inch on either side of strapping			202
Design	Characteristic not in accordance with specified requirements (i.e., required sets of tie tapes missing on the cover)		101	
Material, general	Component not fabricated of the specified material (unless otherwise indicated herein)		102	
Coated cloth (panel, strapping, and patches)	Defect that is visible in this examination shall be classified as listed in 4.4.2.2			
Tie tapes	Cut, frayed, scalloped edge, weak place, or weaving defect: - seriously affecting serviceability		103	
	- affecting serviceability but not seriously			203
Outersole	Defect that is visible in this examination shall be classified as listed in 4.4.2.2			

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TABLE IV. End item visual defects

Examine	Defect	Classification		
		Critical	Major	Minor
Construction and workmanship, general (applicable to all components)	Component or assembly malformed:			
	- seriously affecting serviceability		104	
	- affecting serviceability but not seriously			204
	Any cemented area not fully cured		105	
Repairs	Any unauthorized repair (i.e., coating applied to defect in fabric coating in lieu of patching)	1		
	Edge lifting of patch:			
	- cracking, flaking, or removal of the cement or patch when hand flexed as specified in 3.8	2		
	- edge of patch lifts, but remainder is securely cemented, and will not peel with flexing or moderate pull		106	
	Patch area excessively stiff or tacky affecting serviceability	3		
	Patch less than 1-1/4 inches		107	
	Patch greater than 1-3/4 inches but not greater than 2 inches			205
	Patch greater than 2 inches		108	
	More than four patches per footwear cover		109	
	Heavily coated side of patch cemented to the coated cloth of the footwear cover			206
	Patch extends less than 1/2 inch beyond all edges of the damaged area		110	
	Patch not smoothly and evenly adhered		111	

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TABLE IV. End item visual defects - Continued

Examine	Defect	Classification		
		Critical	Major	Minor
Seams and stitching	Lower tie tape stitching not reinforced or not completely covered by cemented patch		112	
	Seam allowance not maintained, seam edges stitched irregularly or uneven resulting in badly twisted, pleated, or bulging seams but the seam is securely accomplished and sealed			207
	Stitching not securely back-stitched			208
	Not specified seam type		113	
	Not specified stitch type			209
	Stitching omitted where required		114	
	Gage of stitching irregular			210
	Stitching margin (except tie tapes):			
	- under 1/16 inch		115	
	- over 3/16 inch		116	
	- over 1/8 up to 3/16 inch			211
	One or more stitches per inch less than the minimum specified			212
	More than the maximum number of stitches specified not damaging material			213
	Raw edges where not permitted			214
Strapped seams	Open seam resulting in a through hole	4		
	Seam strapped on only one side	5		
	Strapping on both sides in the same area of seam not securely affixed (i.e., strapping not properly cemented so that it can be readily pulled off seam with little resistance)	6		
	Strapping on one side does not completely cover stitching on seam	7		
	Strapping on one side of seam not securely affixed		117	

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TABLE IV. End item visual defects - Continued

Examine	Defect	Classification		
		Critical	Major	Minor
Strapped seams - continued	Blister or delamination in central area of strapping, but edges are securely affixed and the bonded portion of strapping is securely cemented to fabric			215
	Any opening at edge of strapping 1/16 inch or more in depth but does not extend the stitching			216
	Any opening 1/32 inch or more in width extending to the stitching		118	
	NOTE: The extent of opening (or tunnel) may be determined by insertion of any blunt end instrument suitable for this purpose. Penetration of the instrument shall cease immediately if resistance is met.			
	Individual seam not having a continuous length of strapping		119	
	Ends of strapping that meet:			
	- overlapped less than 3/4 inch		120	
	- overlapped more than 1-1/2 inches			217
Marking or label	Missing, incomplete, incorrect, not legible, not specified type or size, not in proper location, or not accomplished in the specified manner			218

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4.4.4 End item dimensional examination. The end items shall be examined for conformance to the dimensions listed below. Any measurement deviating from the specified dimension shall be classified as a defect. The lot size shall be expressed in units of footwear covers. The sample unit shall be one footwear cover. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

All dimensions in 3.6
Strapping width (see 3.3.5)
Overlapping of ends of strapping (see 3.5.5)
Extension of cement line at sides of strapping (see 3.5.5)
Length of bartacking (see 3.5.6.5)
Length of upper tie tapes (see 3.5.6.5)
Length of lower tie tapes (see 3.5.6.6)
Label location (see 3.5.7)

4.4.5 End item testing. The end items shall be tested for the characteristics listed in table V. The lot size shall be expressed in units of footwear covers. The sample unit shall be three completely fabricated footwear covers. The inspection level shall be S-2. Any test failure shall be cause for rejection of the lot.

TABLE V. End item tests

Characteristic	Requirement paragraph	Test method	Number of determinations per sample unit	Results reported as
Initial adhesion of strapping	3.7.1	4.5.1.1	5	Numerically to nearest 0.1 pound per inch of width
Leakage	3.7.2	4.5.4	1	Pass or fail

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4.4.6 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking (exterior and intermediate)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components, such as: incomplete sealing or closure of flap, improper taping, loose strapping, or inadequate stapling Bulged or distorted container
Content	Number of footwear covers per unit pack and intermediate pack is more or less than specified Number of intermediate packs per container is more or less than specified

4.4.7 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirement
Palletization	Pallet pattern not as specified Interlocking of loads not as specified Load not bonded as specified
Weight	Exceeds maximum load limits
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

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4.5 Methods of inspection.

4.5.1 Adhesion of strapping tests.

4.5.1.1 Initial adhesion. After the specimen has aged 48 hours, the adhesion of strapping shall be determined as specified in Method 5962 of FED-STD-191, except that adhesion shall be the average of the three lowest individual load resistance registered on the autographic chart divided by the width of the seam. Any separation of the coating in lieu of strapping adhesion will be considered an invalid test and a new specimen will have to be tested.

4.5.1.2 Adhesion after water immersion. The seams shall be immersed for 2 hours in boiling water, removed, immersed for 15 minutes in water at 75° ± 5°F, removed from the water, and, while still dripping wet, tested for strapping adhesion as specified in 4.5.1.1.

4.5.1.3 Adhesion after heat aging. The specimen shall be exposed for 7 days in a circulating air oven at 158° ± 2°F, removed from the oven, conditioned for 24 hours at 70° ± 2°F and 64 ± 2 percent relative humidity, and then tested for strapping adhesion as specified in 4.5.1.1.

4.5.2 Cold crack of cement film test. Test specimens of the coated cloth specified in 3.3.1, measuring 8 by 8 inches minimum, shall be cleaned on one side only by wiping with a suitable cleaning agent that is not a solvent for butyl coating. The cleaned area shall then be coated with two layers of the cement specified in 3.3.6. Test specimens shall then be conditioned for 4 hours at room temperature and dusted until tack free.

4.5.2.1 Procedure. Specimens shall be tested as specified in Method 5874 of FED-STD-191 except that testing temperature shall be minus 20°F and time of exposure shall be for a maximum of 4 hours before testing. When performing tests, the cemented side shall be considered the heavily coated side.

4.5.3 Hydrostatic resistance test. Ten 8 inch by 8 inch specimens shall be randomly selected from a 1 yard piece of the coated cloth specified in 3.3.1. Five specimens shall be tested for hydrostatic resistance as specified in Method 5512 of FED-STD-191. The remaining five specimens shall be cement coated as specified in 4.5.2 and tested for hydrostatic resistance in accordance with Method 5512. The sides without the cement coating shall be exposed to the water.

4.5.4 Leakage test. The footwear cover shall be connected to a regulated source of compressed air by means of a suitable adapter at the open end. A differential air pressure equivalent to a head of 35 ± 5.0 mm of water shall be maintained inside the cover for 1 minute while a 0.25 percent mild soap solution is applied over all exterior seams. The appearance of bubbles shall be regarded as evidence of leakage. The covers shall be dried and retalced prior to packaging.

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end of the container with the heel toward the front and the sole facing

4.5.5 Resistance of label to fading and tackiness.

4.5.5.1 Boiling water test. Completely submerge the labels in boiling water for a period of 1 hour. Remove and examine for legibility and tackiness.

4.5.5.2 Acetic acid test. Completely submerge the labels in 10 percent (by volume) acetic acid solution at 75° to 80°F for 1 hour. Remove and examine for legibility and tackiness.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit packing. Each cover shall be neatly folded with the tie tapes inside the folds, to measure approximately 16 by 6-1/2 inches. Two folded covers, of one size only, placed together heel to toe shall be inserted in a snug-fitting flat or square style polyethylene film bag conforming to A-A-50083.

5.1.1.2 Intermediate packs. Six covers (three pairs), of one size only, unit packed as specified in 5.1.1.1, shall be placed flat in an intermediate fiberboard box conforming to style RSC, type CF, variety SW, class domestic, grade 275 of PPP-B-636. Inside dimensions of the fiberboard box shall be 16 inches in length, 6-1/2 inches in width, and 7-1/2 inches in depth. Box closure shall be secured with 2 inch minimum width gummed paper tape conforming to type III, grade B of PPP-T-45.

5.1.2 Commercial. Covers shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial, as specified (see 6.2).

5.2.1 Level A packing. Forty-eight covers (twenty-four pairs) of one size only, preserved as specified in 5.1, shall be packed in a fiberboard shipping container conforming to style RSC, grade V2s of PPP-B-636. Level A intermediate packs shall be packed flat two in length, two in width, and two in depth within a shipping container. Inside dimensions of each shipping container shall approximate 33 inches in length, 14 inches in width, and 16 inches in depth. Approximate dimensions are furnished as a guide only. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.6. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.2). Strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

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5.2.2 Level B packing. Forty-eight covers (twenty-four pairs) of one size only, preserved as specified in 5.1, shall be packed in a fiberboard shipping container conforming to style RSC, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. Level A intermediate packs shall be packed flat two in length, two in width, and two in depth within a shipping container. Inside dimensions of each shipping container shall approximate 33 inches in length, 14 inches in width, and 16 inches in depth. Approximate dimensions are furnished as a guide only. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.6.

5.2.2.1 Weather-resistant fiberboard containers. When specified (see 6.2), the shipping container shall be grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.6.

5.2.3 Commercial packing. Covers, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Palletization. When specified (see 6.2), covers, packed as specified in 5.2.2 and 5.2.3, shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Pallet type shall be type I (4-way entry), type IV, or type V in accordance with MIL-STD-147. Pallets shall be fabricated from wood groups I, II, III, or IV of MIL-STD-731. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L or film bonding means O or P. Pallet pattern shall be number 2 in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course.

5.4 Marking. In addition to any special marking required by the contract or purchase order, intermediate packs, shipping containers, and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The footwear cover is intended for use over regular or protective footwear in conjunction with the M-3 Toxicological Agents Protective Coverall. Footwear covers are to be used in pairs; however, there is no distinction between left and right. All footwear covers in the same size are interchangeable. The cover will fit over combat, hip, and knee boots as follows:

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Boot sizes	Footwear cover sizes
Up to 6-1/2	Small
7 to 11-1/2	Medium
12 and over	Large

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Size required (see 1.2 and 6.1).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When a first article is required (see 3.1, 4.3, and 6.3).
e. Levels of preservation and packing (see 5.1 and 5.2).
f. Type and class of unit load required (see 5.2.1).
g. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
h. When palletization is required (see 5.3).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Sample. For access to samples, address the contracting activity issuing the invitation for bids (see 3.2).

6.5 Subject term (key word) listing.

Chemical protection

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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Air Force - 99

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Air Force - 82
DLA - CT

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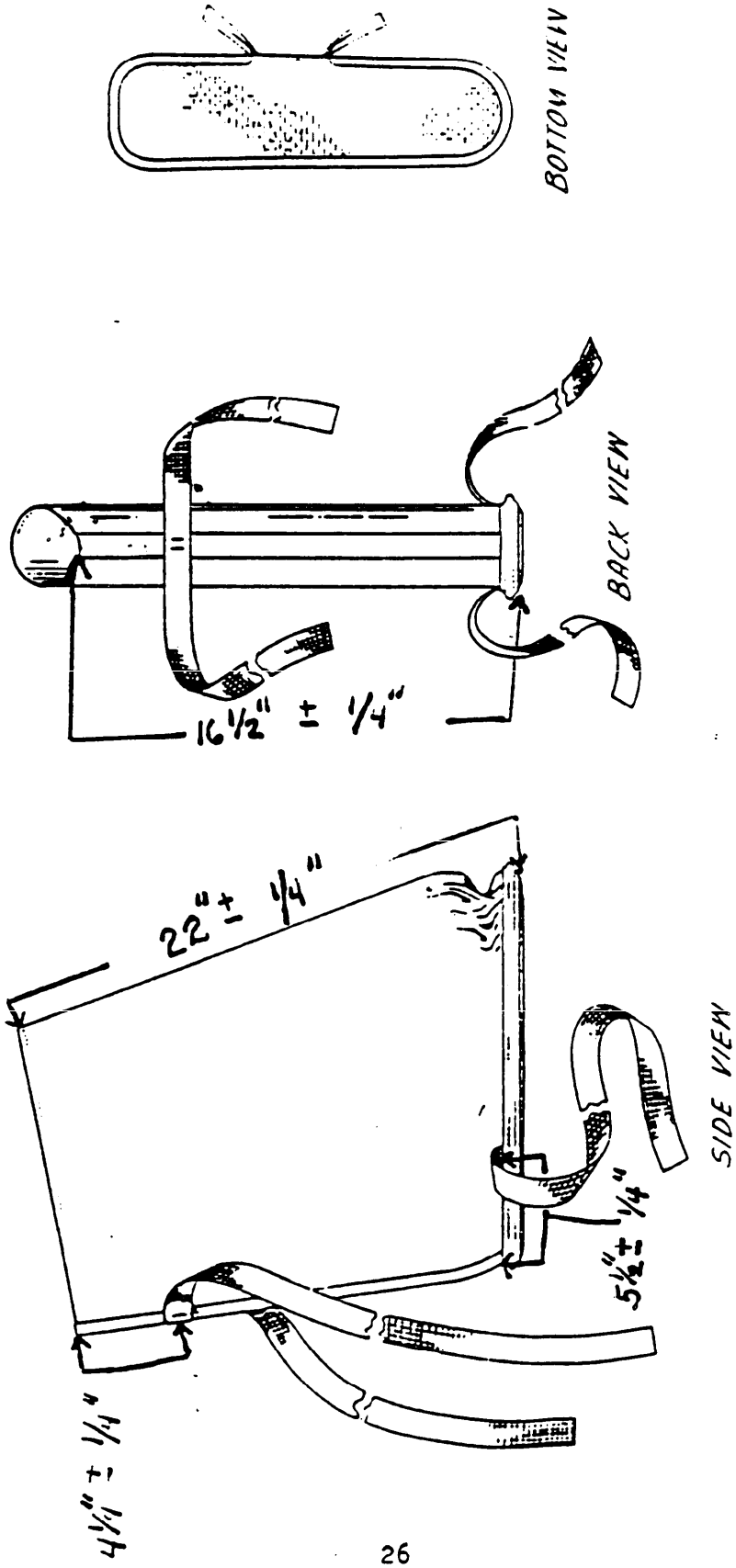


Figure 1. Footwear Covers, Toxicological Agents Protective, M-1

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-F-12224J	Footwear Covers, Toxicological Agents Protective, M-1

3. NAME OF SUBMITTING ORGANIZATION

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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